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SECTORAL LABOUR SUPPLY, CHOICE RESTRICTIONS AND FUNCTIONAL FORM

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SUMMARY

In this paper we discuss a general framework for analysing labour supply behaviour in the presence of complicated budget and quantity constraints, of which some are unobserved. The individual's labour supply decision is viewed as a choice from a set of discrete alternatives (jobs). These jobs are characterized by attributes such as hours of work, sector-specific wages and other sector-specific aspects of the jobs.

The labour supply model for married women is estimated on Norwegian data. Wage elasticities and the outcome of a tax reform experiment show that overall labour supply is moderately elastic, but these modest overall responses shadow for much stronger inter-sectoral changes. Our model is compared with a discrete choice model in which the utility is assumed to be a polynomial. We show that our estimated model has a more economically sensible interpretation and fits the data as well as the alternative approach. Copyright © 2006 John Wiley & Sons, Ltd.

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1. INTRODUCTION

In this paper we discuss a particular framework for modelling labour supply behaviour in the presence of complicated budget sets, qualitative job attributes and restrictions on hours of work, and apply this framework to analyse data on workers' observed choice of sector and hours in the labour market. Compared to earlier attempts, our framework for estimating labour supply allows for a more complete empirical strategy in circumstances where job choices are not adequately represented by hours and wages.

In most western countries the tax and social benefit rules imply a nonlinear tax and benefit system with effective marginal tax rates that do not increase uniformly with income. The latter implies a non-convex budget set. Fixed cost of working and tax deductions if working contribute to these non-convexities. In recent years the tax and benefit system has been simplified in many countries, but there are still non-convexities present in most budget sets. Burtless and Hausman (1978) were the first to propose a modelling approach that explicitly accounted for the entire budget set generated by a nonlinear tax system, which could even imply a non-convex budget set. To account for nonlinear and possibly non-convex budget sets by using the Hausman approach becomes rather cumbersome when there is more than one adult in the household, or when complicated social benefit and tax deduction rules are present. Moreover, in order to ensure convex preferences at kink points of the budget set the Hausman approach yields additional ad hoc restrictions on the model beyond

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those that follow from economic theory, see MaCurdy *et al.* (1990), MaCurdy (1992) and Heim and Meyer (2003).¹

In contrast, the particular approach advocated in our paper, and which we shall describe in a moment, has the advantage that it becomes simple to handle complicated nonlinear tax and other non-standard choice restrictions. This is also the case for two-person households.

In most previous empirical labour supply studies the mathematical structure of the modelling framework rests upon the assumption that the fundamental choice variables of the household are 'consumption' (composite) and 'leisure' (hours of work), which can be chosen freely subject to the economic budget constraint. Yet, it seems apparent that hours of work and income are only two out of several job-related attributes, which are important for individual behaviour in the labour market. 'Type of work', and other 'non-pecuniary job attributes', often matter a great deal, and may even be more important than hours of work. An extreme example of the latter phenomenon is found among scientists, artists and government bureaucrats for whom specific work activities represent major means for self-realization. Another characteristic of the labour market is that hours of work are fixed for many types of job. Thus, if an individual wishes to change his or her workload he or she would in this case have to change job.² This property is consistent with the findings of Altonji and Paxson (1988).

In view of the arguments above it may be more appropriate to consider labour supply behaviour as the outcome of households choosing from a finite set of job 'packages', each of which is characterized by an offered wage rate, offered hours of work and non-pecuniary attributes. The individual-specific choice sets of job opportunities may be thought of as being determined by employers or in negotiation between employers and unions, and they are thus exogenous to the individuals. The qualitative job attributes are often unobservable, or at most only partly observable to the analyst. This is the point of departure taken in this paper. Specifically, the choice environment is assumed to consist of a latent, individual-specific set of jobs. A job is characterized by a combination of fixed hours of work, wage rate and non-pecuniary job attributes (such as type of work and working conditions). The notion of individual-specific choice sets is important for our modelling of choice constraints. In our setup there are thus two sources of unobserved heterogeneity, unobserved heterogeneity in tastes and in opportunities.

In Dagsvik (1994) a general framework for modelling this type of setting was developed. Simplified versions of this framework have been applied by Anderson *et al.* (1988), Aaberge *et al.* (1990, 1995, 1999) and Dagsvik and Strøm (1997) to analyse labour supply behaviour. In contrast, our paper is more theoretical in that it focuses on a detailed discussion and interpretation of how functional form and the probability law of unobservables can be justified from behavioural arguments, and we investigate empirically the properties of alternative functional form specifications.

Previous attempts to take (quantity) constraints on the choice set into account have been restricted solely to one job attribute, namely hours of work. Contributions by Ilmakunnas and Pudney (1990), Kapteyn *et al.* (1990), Dickens and Lundberg (1993) and van Soest (1994) emphasize the inability of standard empirical labour supply models to account for observed peaks in the hours of work distribution at part-time and full-time hours. They have discussed approaches

¹ The 'Hausman approach' in estimating labour supply introduced in 1978 initiated numerous empirical studies on labour supply using variants of this approach; Blomquist (1983, 1992), Arrufat and Zabalza (1986), Hausman (1980, 1981, 1985), Hausman and Ruud (1984). For surveys of studies using the 'Hausman approach', see Blundell and MaCurdy (1999).

² Alternatively, the worker may have to change the content of his current job.

to take account of this type of constraint in the econometric modelling of labour supply. These approaches are, however, different from the one developed in our paper. In Dagsvik and Strøm (2004), we show that these previous studies can be viewed as a special case of our model.

In all of these recent labour supply contributions the individuals are assumed to have the same wage across jobs. Thus, in previous labour supply studies it is assumed that an individual has a fixed wage rate, and the possibility of job-specific wages is ignored. Labour market theories, like the theories of efficient wages and trade unions, suggest that wages may differ across jobs, see for example Krueger and Summers (1988) and Edin and Zetterberg (1992). In contrast, our framework can allow for job-specific wage rates. In the labour supply literature there are also approaches that allow offered job-specific wage rates to vary systematically with hours worked (Moffitt, 1984), or effort decisions (Copeland and Monnet, 2004). In our data we observe annual hours and annual wage income and we obtain wage rates through division. These computed wage rates may include overtime pay, but in the highly unionized and government-regulated Norwegian economy, overtime pay plays a minor role in the wage contracts.

The empirical part of the paper deals with labour supply among married females in Norway in 1994, who face choice sets of feasible jobs within the public and the private sector of the economy. The estimated model is used to simulate the impact on labour supply of introducing a flat tax reform, given that tax revenue is kept constant. Both the wage elasticities and the tax reform analysis indicate that overall labour supply is moderately elastic, but these moderate responses shadow for much stronger inter-sectoral responses.

Recently, van Soest *et al.* (2002) have proposed a labour supply model where the agent faces no restrictions other than the budget constraint, and the 'menu' of feasible hours of work is discrete. Moreover, they represent the structural part of the utility by a polynomial in hours and consumption. We use our data to estimate several versions of the model proposed by van Soest *et al.* (2002), and make a detailed comparison with our model. It turns out that our framework yields a more economically sensible interpretation in terms of convexity of preferences, and also implies goodness of fit that is at least as satisfactory as the alternative approach.

2. THE MODELLING FRAMEWORK

As alluded to above, the choice environment of a worker is assumed to consist of a set of latent job and non-market opportunities. Each job is characterized by fixed observed attribute variables that represent the contract (wage rate and hours of work) and unobserved attributes that describe the job type. We shall first discuss the case where qualitative attributes are latent. Later, we extend the framework to accommodate sector-specific jobs (public sector versus private sector), where choice of sector is observed.

2.1. Preferences and Choice Sets

Let $U(C, h, z)$ be the utility function of the household where C denotes household consumption, z indexes the market and non-market opportunities, or job types, and h is hours of work of the married female. Let positive indices, $z = 1, 2, \dots$, refer to market opportunities (jobs) and $z = 0$ refer to the non-market alternative. To a market opportunity z , there are associated hours of work, $H(z)$ and wage rate, $W(z)$. How these are determined will be discussed later. The opportunity index z in the utility function accommodates the notion that workers may have preferences over

job types (which includes preferences for working in specific sectors of the economy) in addition to income and hours of work. For given hours and wage rate, h and w , the economic budget constraint is represented by

$$C = f(hw, I) \quad (1)$$

where I is non-labour income, which includes the income of the husband, and $f(\cdot)$ is the function that transforms gross income into after-tax household income. Here, the income of the husband is treated as given. The function $f(\cdot)$ captures all details of the tax and benefit system. The price index for the composite good, C , is set equal to one. Our first assumption concerns the structure of the preferences. The utility function is assumed to have the structure

$$U(C, h, z) = v(C, h)\varepsilon(z) \quad (2)$$

for $z = 0, 1, 2, \dots$, where $v(\cdot)$ is a positive deterministic function and $\varepsilon(z)$ is a positive random taste-shifter. However, the separability assumption in (2) will be relaxed in the empirical application below through the introduction of random effects in the systematic term $v(C, h)$.³ The random taste-shifter is assumed to account for the unobservable individual characteristics and non-pecuniary job-type attributes that affect utility. For notational simplicity we will use the notation

$$\psi(h, w; I) \equiv v(f(hw, I), h) \quad (3)$$

The term $\psi(h, w; I)$ is the representative utility of jobs with hours of work h and wage rate w , given non-labour income I . In addition to (1), there are restrictions on the set of feasible market opportunities a specific worker faces because there are job types for which the worker is not qualified and there may not be jobs available for which she is qualified.

2.2. A Simplified Description of the Model

Since the general model is based on a somewhat abstract representation of the unobserved heterogeneity in choice restrictions and preferences, we shall first consider a simplified version. In this simplified version unobserved heterogeneity in the choice sets is ignored. Furthermore, it is assumed that the agent can only choose among a finite set of jobs. Let $D(h, w)$ denote the agent's set of available jobs with hours of work, $H(z) = h$ and wage rate, $W(z) = w$. Let $m(h, w)$ be the number of jobs in $D(h, w)$. For the non-market alternative one can normalize such that $m(0, 0) = 1$. The random error terms $\{\varepsilon(z)\}$ are assumed independent of offered hours and wages.

We next assume that the choice of a job, or the non-market alternative from a given set of alternatives, satisfies the Independence from Irrelevant Alternatives (IIA) property, see Luce (1959). Recall that the underlying intuition of the IIA assumption is, loosely speaking, that the agent's ranking of job opportunities from a subset B (say), within the choice set of feasible jobs with given job-specific hours of work and wage rates, does not change if the choice set of feasible jobs is altered. Recall also that the stochastic formulation of IIA adopted by the psychologists, means that this property is only claimed to hold *on average*, for an agent that is exposed to a large number of repeated choice experiments, cf. Thurstone (1927). In appendix A in Dagsvik and

³ Let η denote the random effects in the systematic term and let $\bar{v}(C, h) = E v(C, h, \eta)$. Then we can rewrite $v(C, h, \eta)\varepsilon(z)$ as $\bar{v}(C, h)\varepsilon^*(C, h, z)$, where $\varepsilon^*(C, h, z) = (v(C, h, \eta)/\bar{v}(C, h))\varepsilon(z)$ is the new taste-shifter that captures the random effect. We notice that $\varepsilon^*(C, h, z)$ depends on C and h in addition to z .

Strøm (2004) we discuss further interpretations of IIA. As mentioned above, we shall allow for random effects, which means that in the empirical application below IIA is only assumed to hold conditional on the random effect.⁴

Consider now the choice model. As is well known (Strauss, 1979), IIA implies that the random error terms $\{\varepsilon(z)\}$ can, with no loss of generality, be assumed independent with c.d.f. $\exp(-x^{-1})$, $x > 0$.⁵ Note that the agent will be observed working h hours at wage rate w if she chooses a job within $D(h, w)$. Let $\varphi(h, w; I)$ denote the probability that the agent shall choose a particular job with offered hours h and wage rate w , when her non-labour income is I . From standard results in discrete choice theory (McFadden, 1984) it now follows that the probability that a *specific* job z (say) within $D(h, w)$ shall be the feasible job with the highest utility equals

$$P\left(\psi(h, w; I)\varepsilon(z) = \max_{x, y} \max_{k \in D(x, y)} (\psi(x, y; I)\varepsilon(k))\right) = \frac{\psi(h, w; I)}{\sum_{x, y} \sum_{k \in D(x, y)} \psi(x, y; I)} = \frac{\psi(h, w; I)}{\sum_{x, y} \psi(x, y; I)m(x, y)}$$

The probability of choosing *any* job within $D(h, w)$ is thus obtained by summing over all jobs in $D(h, w)$, which yields

$$\begin{aligned} \varphi(h, w; I) &= \sum_{z \in D(h, w)} P(\psi(h, w; I)\varepsilon(z)) = \max_{x, y} \max_{k \in D(x, y)} (\psi(x, y; I)\varepsilon(k)) \\ &= \sum_{z \in D(h, w)} \frac{\psi(h, w; I)}{\sum_{x, y} \sum_{z \in D(x, y)} \psi(x, y; I)} = \frac{\psi(h, w; I)m(h, w)}{\sum_{x, y} \psi(x, y; I)m(x, y)} \end{aligned} \quad (4)$$

The resulting expression is a choice model that is analogous to a multinomial logit model with representative utility terms $\{\psi(h, w; I)\}$ weighted with the frequencies of feasible jobs, $\{m(h, w)\}$. Unfortunately, the frequencies $\{m(h, w)\}$ are not directly observable, but under specific assumptions one can identify $m(h, w)$ and $\psi(h, w; I)$ and estimate their parameters. We shall return to this issue below.

2.3. The General Case with Stochastic Choice Sets

In the simplified model version described above the choice sets, represented by $\{D(h, w)\}$, were assumed to be equal across observationally identical agents. It is, however, desirable to allow choice sets also to vary across agents due to unobserved heterogeneity in opportunities. In this

⁴ A famous example often used to demonstrate how restrictive IIA is, is the so-called red bus–blue bus example. However, this example is rather extreme because two of the alternatives are practically identical. In most cases alternatives are different in a more essential way. It is true though that the alternatives in many cases can differ in varying degrees with respect to common aspects, which may weaken the plausibility of IIA. Nevertheless, it is still not evident that different degrees of substitutability between alternatives will yield violation of IIA on the individual level. If the stochastic elements of the utility function to a large extent represent bounded rationality, they can be viewed as pure ‘noise’, and therefore may not reflect substitution relations in a systematic way.

⁵ The c.d.f. $\exp(-x^{-1})$, $x > 0$, occurs in this context because of the multiplicative specification of the utility function in assumption 1. This c.d.f. is completely equivalent to the more familiar c.d.f. $\exp(-e^{-x})$ that occurs in the corresponding additive formulation. Recall that the standard type I distribution function has the form $\exp(-1/y)$, $y > 0$, cf. Resnick (1987). In the statistical literature there is some confusion, with some authors calling this distribution type III.

subsection we shall present a modelling framework that accommodates the notion of both observed and unobserved heterogeneity in opportunities and preferences.

To this end we turn to discuss a particular representation of random choice sets through the set of triples $\{(H(z), W(z), \varepsilon(z)), z = 1, 2, \dots\}$ associated with the respective jobs. We shall assume that these triples are independent and randomly scattered in some suitable set Ω . This conforms with the intuition that due to the researcher's lack of information, the location of points in Ω appears to be independent. A convenient formal representation of this type of stochastic 'process' is obtained through the multidimensional non-homogeneous spatial Poisson process. Recall that a Poisson process on Ω has the property that its points are randomly scattered throughout Ω . If the Poisson process is homogeneous, the points are randomly but evenly dispersed on Ω . In contrast, the non-homogeneous Poisson process allows for uneven distribution of points in the sense that there are, on average, higher concentrations of points in some parts of Ω than in others. This allows us to account for the possibility that there may be fewer jobs available with high values of the taste shifters $\{\varepsilon(z)\}$ than jobs with low values of $\{\varepsilon(z)\}$, and the possibility that there may be more jobs with full-time hours available than jobs with other hours of work. To facilitate exposition it may be convenient to explain the multidimensional Poisson process representation by first starting with the more familiar one-dimensional Poisson process, namely the one that represents the taste-shifters $\{\varepsilon(z)\}$. Subsequently, we extend the process to higher dimensions by 'assigning' further attributes (job-specific hours and wage rates) to the one-dimensional Poisson points. Thus, we assume that the taste-shifters $\{\varepsilon(z), z = 1, 2, \dots\}$ are realizations of a non-homogeneous Poisson process on $(0, \infty)$ with intensity $\kappa(\varepsilon)$. Moreover, $\{\varepsilon(z), z = 1, 2, \dots\}$ are distributed independently of $\varepsilon(0)$. As mentioned above, the points (taste-shifters) of the Poisson process are independent and randomly scattered, and the probability that there is a job with $\varepsilon(z) \in (\varepsilon, \varepsilon + d\varepsilon)$ available to the individual equals $\kappa(\varepsilon)d\varepsilon$. Moreover, the probability that there is more than one job with taste-shifters within $(\varepsilon, \varepsilon + d\varepsilon)$ is negligible. Finally, we assume that hours of work and wage rates $(H(z), W(z)), z = 1, 2, \dots$, are i.i.d. with joint density $g(h, w)$, where $0 < h < \bar{h}$, and $\bar{h}$ is an upper bound on hours of work. Moreover, $\{(H(z), W(z)), z = 1, 2, \dots\}$ are independent of $\{\varepsilon(z), z = 0, 1, 2, \dots\}$.

From these assumptions we realize that the points $\{H(z), W(z), \varepsilon(z)\}$ can be interpreted as realizations from a three-dimensional non-homogeneous Poisson process on Ω with intensity $g(h, w)\kappa(\varepsilon)$. A formal proof of this is found in Resnick (1987, p. 135). This intensity provides a complete representation of the Poisson process in the sense that it governs the corresponding probability distribution of the Poisson points in Ω .

There are two different interpretations of the notion of random choice sets. In the first interpretation the randomness is viewed as a result of unobserved heterogeneity in choice opportunities across the population. While the individual agent is viewed as having complete information about her choice set and the utilities of every feasible alternative, the researcher cannot observe this heterogeneity. He can at most account for heterogeneity in the choice set due to observed individual characteristics. Given the set of feasible jobs, one can interpret $g(h, w)dhdw$ as the probability (from the researcher's point of view) that, for a randomly selected agent, there is a job with hours of work $H(z) \in (h, h + dh)$ and $W(z) \in (w, w + dw)$ that is feasible. Loosely speaking, the corresponding 'empirical' counterpart is the fraction of the feasible jobs that have hours of work and wage rates within $(h, h + dh) \times (w, w + dw)$.

The second interpretation is related to bounded rationality in the sense that the agent has difficulties with assessing the precise value to him of the respective alternatives and also has limited capacity to identify and take into account the exact choice set.

From the assumptions above it is now possible to derive choice probabilities of observed labour market choices. However, without further restrictions on the intensity of the Poisson process the expression for the choice probabilities will be rather general and complicated analytically. It is thus desirable to add further behavioural assumptions to restrict the modelling framework. Similarly to the simplified setup in Section 2.2, we shall continue to assume that IIA holds.

We are now ready to express the probability distribution of *realized* hours and wages, including the probability of not working. Let $\Phi(h, w; I)$ be the joint cumulative distribution of realized hours and wages that follow from utility maximizing behaviour, i.e.

$$\Phi(h, w; I) \equiv P \left(\max_{H(z) \leq h, W(z) \leq w} (\psi(H(z), W(z); I)\varepsilon(z)) = \max_z (\psi(H(z), W(z); I)\varepsilon(z)) \right) \quad (5)$$

Equation (5) defines the probability that the chosen opportunity (i.e. job) has hours of work less than or equal to h and wage rate less than or equal to w . Under IIA and the assumptions above, and provided $\kappa(\varepsilon)$ is continuous and the integral of $\kappa(x)$ over $[x, \infty)$ is finite for $x > 0$, one can show that $\varepsilon(0)$ is type I extreme value distributed and the intensity $\kappa(\varepsilon)$ has the form $\theta\varepsilon^{-2}$, for positive ε , where θ is a positive constant. Moreover, the probability density $\varphi(h, w; I)$ is given by

$$\varphi(h, w; I) = \frac{\psi(h, w; I)g(h, w)\theta}{\psi(0, 0; I) + \theta \int \int_D \psi(x, y; I)g(x, y)dx dy} \quad (6)$$

for $h > 0, w > 0$ and

$$\varphi(0, 0; I) = \frac{\psi(0, 0; I)}{\psi(0, 0; I) + \theta \int \int_D \psi(x, y; I)g(x, y)dx dy} \quad (7)$$

for $h = w = 0$, where $D = [0, \bar{h}] \times (0, \infty)$. The proof of this result is given in appendix B of Dagsvik and Strøm (2004), and it is an adaptation of a more general result in Dagsvik (1994). Recall that when $\varepsilon(0)$ is type I extreme value distributed the c.d.f. is given by $\exp(-1/x)$ for positive x . The parameter θ that appears in the intensity $\kappa(\varepsilon)$ is a measure of job availability, since a high value of θ means that there is a high probability that there is an available job with taste-shifters belonging to $(\varepsilon, \varepsilon + d\varepsilon)$ for any positive ε . In Dagsvik and Strøm (2004) we provide some intuition why $\theta \leq 1$.

Although we have assumed that the agent's taste-shifters are (stochastically) independent of offered hours and wage rates, the opportunity density will depend on the *distribution* of the preferences due to equilibrium conditions. In other words, the market forces that regulate the balance between supply and demand, be it a market clearing regime or not, are assumed to operate solely on an aggregate level. The opportunity density will consequently depend on the production technology of the firms as well as on the contract and wage-setting policies of the unions and the firms. It is beyond the scope of this paper to discuss fully how the opportunity density $\theta g(\cdot)$, through market equilibrium processes, depends on the systematic part of the utility function, $\psi(\cdot)$. This means that the estimated model can only be applied to simulate behaviour conditional on the opportunity density. This parallels the assumptions made in previous labour supply models, in which it is assumed that the agents take wages as given and that the agent can freely choose between any hours of work. In Dagsvik (2000), it is discussed how an explicit

equilibrium model version can be specified and how the opportunity density depends on workers' preferences and firms' technologies, and one can use that framework to justify that the opportunity density is multiplicatively separable in hours and wage rates, which means that the offered hours and wage rates are independent. Specifically, in appendix C of Dagsvik and Strøm (2004) we show how this property follows from the setup in Dagsvik (2000), combined with the assumption of constant returns to scale production functions of the firms. Another type of justification for this assumption is that offered wages, in the unionized part of the economy, are set in yearly wage settlements. Normal working hours, on the other hand, are determined more infrequently, typically once or twice every decade.

Identification of the model given in (6) and (7) was considered in Dagsvik and Strøm (1997) under the assumption that the structural term of the utility function is multiplicatively separable in consumption and hours of work (leisure). However, here the assumption of a multiplicatively separable utility function will be relaxed. With reference to the arguments given in appendix C of Dagsvik and Strøm (2004), we have proved identification (Dagsvik and Strøm, 2004) when

$$g(h, w) = g_1(w)g_2(h) \quad (8)$$

which means that hours and wage rates, $\{H(z)\}$ and $\{W(z)\}$, offered by the firms are independent.

2.4. Functional Form

Current quantitative economic research often suffers from the lack of theoretical principles on which assumptions about functional form can be made. While elaborate and sophisticated theoretical models of behaviour exist, such models are often not detailed enough to be useful for purposes other than qualitative predictions. The standard approach in empirical analyses is to 'let the data determine' functional forms within ad hoc selected parametric classes (including so-called flexible functional forms). This is clearly unsatisfactory in the context of structural modelling. We shall now apply a typical approach within the field of psychophysics, as presented in Falmagne (1985), to restrict the class of functional forms for the systematic part of the utility function.

To this end, consider the particular case with an opportunity distribution that has all mass in two points (h, w_1) and (h, w_2) , with probability mass equal to 0.5 in either point. [Since preferences are assumed independent of opportunities, the analyst is, for the sake of interpretation and theoretical analysis, free to select any opportunity distribution he finds suitable for a specific purpose while keeping the function $v(\cdot)$ unchanged.] Let L denote leisure defined as total time available for work minus hours of work. Then from (6) it follows that we can write

$$\frac{v(C_1, h_1)}{v(C_1, h_1) + v(C_2, h_2)} \equiv \tilde{\varphi}(C_1, L_1; C_2, L_2) \quad (9)$$

where $\tilde{\varphi}(C_1, L_1; C_2, L_2)$ is the probability that (C_1, L_1) is preferred to (C_2, L_2) . It is understood that consumption in this context means disposable income minus subsistence expenditure and leisure means leisure minus subsistence leisure. Assume now that (i) whenever $C_1, C_2, C_1^*, C_2^*, L_1, L_2$, are such that

$$\tilde{\varphi}(C_1, L_1; C_2, L_2) \leq \tilde{\varphi}(C_1^*, L_1; C_2^*, L_2) \text{ then } \tilde{\varphi}(rC_1, L_1; rC_2, L_2) \leq \tilde{\varphi}(rC_1^*, L_1; rC_2^*, L_2)$$

for any positive r . Furthermore, assume that (ii) whenever $L_1, L_2, L_1^*, L_2^*, C_1, C_2$, are such that

$$\tilde{\varphi}(C_1, L_1; C_2, L_2) \leq \tilde{\varphi}(C_1, L_1^*; C_2, L_2^*) \text{ then } \tilde{\varphi}(C_1, rL_1; C_2, rL_2) \leq \tilde{\varphi}(C_1, rL_1^*; C_2, rL_2^*)$$

for any positive r . Assumption (i) states that if the fraction of workers who prefer jobs that yield (C_1, L_1) to jobs that yield (C_2, L_2) is less than the fraction of workers who prefer jobs that yield (C_1^*, L_1) to jobs that yield (C_2^*, L_2) , then the same is true when the respective consumption levels are scale transformations of the original levels. Recall that the non-pecuniary characteristics of the jobs are represented by random terms that are independent of the systematic terms $\{v(C, h)\}$. Note that assumption (i) does *not* claim that $\tilde{\varphi}(rC_1, L_1; rC_2, L_2)$ is independent of r . It only expresses that if the number of individuals who prefer (C_1^*, L_1) to (C_2^*, L_2) is greater than the number of individuals who prefer (C_1, L_1) to (C_2, L_2) , this inequality remains true when consumption levels are increased or decreased by the same factor. For the sake of understanding the limitation of assumption (i), we can think of two objections against this assumption. One objection is that the individual's perception about his personal subsistence level may be somewhat vague and may not be identified by a single fixed amount. Rather it may vary from one moment to the next according to fluctuations in his mood and state of mind. Another objection is related to satiation. If satiation is present and rC_1^* and rC_2^* are close to satiation levels for sufficiently large r and $L_1 = L_2 = L$ (which means that the deterministic part of the utility approaches a constant), the second inequality in assumption (i) may be reversed because $\tilde{\varphi}(rC_1^*, L; rC_2^*, L)$ will be close to 0.5, independent of the levels of C_1^*, C_2^* and L . In the absence of satiation it seems to us to be highly unreasonable that the inequality $\tilde{\varphi}(C_1, L_1; C_1, L_2) \leq \tilde{\varphi}(C_1^*, L_1; C_2^*, L_2)$ should be reversed if the consumption levels are rescaled, i.e. that assumption (i) is violated.

We realize that assumption (ii) is completely analogous to assumption (i) and thus the motivation is similar.

In Dagsvik and Strøm (2004), appendix B, it is proved that if (9) and assumptions (i) and (ii) hold with $v(C, h)$ continuous and strictly increasing in C and strictly decreasing in h , then

$$\log v(C, h) = \beta_1 \frac{(C^{\alpha_1} - 1)}{\alpha_1} + \beta_2 \frac{(L^{\alpha_2} - 1)}{\alpha_2} + \beta_3 \frac{(C^{\alpha_1} - 1)(L^{\alpha_2} - 1)}{\alpha_1 \alpha_2} \quad (10)$$

where $\{\alpha_j\}$ and $\{\beta_j\}$ are constants with $\beta_j > 0, j = 1, 2$, and β_3 can be negative. If one imposes the stronger assumption that $\tilde{\varphi}(rC_1, L_1; rC_2, L_2)$ is *independent* of $r > 0$, it can easily be demonstrated that this implies that $\alpha_1 = 0$. If one imposes the assumption that $\tilde{\varphi}(rC_1, sL_1; rC_2, sL_2)$ is *independent* of r and s , for all $r > 0, s > 0$, it is easily seen that this yields that $\alpha_1 = \alpha_2 = \beta_3 = 0$ so that (10) reduces to

$$\log v(C, h) = \beta_1 \log C + \beta_2 \log L \quad (11)$$

which is the well-known Stone Geary utility function. (Recall that C and L are defined as consumption and leisure minus the respective subsistence levels.)

The result in (10) does not depend on the particular form of $\tilde{\varphi}(\cdot)$ given in (9). In fact, it can easily be verified that it is enough to assume that $\tilde{\varphi}$ has the form

$$\tilde{\varphi}(C_1, L_1; C_2, L_2) = F \left(\frac{v(C_1, h_1)}{v(C_2, h_2)} \right) \quad (12)$$

where F is any strictly increasing c.d.f. on R_+ . This is so because the proof does not depend on the form of $F(\cdot)$.

2.5. Extension to Several Sectors

An essential motivation for the framework discussed in this paper is that it is particularly convenient for modelling workers' choices among jobs with observable non-pecuniary job attributes. In general, jobs in different sectors may differ with respect to non-pecuniary attributes, such as job security (with the government sector at one extreme and private export industries at the other) and the nature of the tasks to be performed.

In this section we shall outline how the model can be extended to a multi-sectoral setting. To this end, we now suppose that the agent can choose among n sectors. The utility function in this case is assumed to have the structure

$$U(C, h, j, z) = v_j(C, h)\varepsilon_j(z) \quad (13)$$

for $j = 1, 2, \dots, n$, where j indexes sector, $z = 0, 1, \dots$, and $j = 0, z = 0$, represents 'not working', and $v_j(C, h)$ is the representative utility of consumption and hours of work (C, h) and working in sector j while $\{\varepsilon_j(z)\}$ are the corresponding error terms. Now (8) is extended to involve sector-specific densities $g_j(h, w)$ where $g_j(h, w)$ represents the joint opportunity density of the feasible hours of work and wage rates in sector j . Also the Poisson process representation of the job opportunities is allowed to be sector-specific, with sector-specific parameter θ_j associated with the intensity of the taste-shifters of the Poisson processes. Thus, $g_j(h, w)dh dw$ is the mean fraction of feasible jobs in sector j with offered hours of work and wage rates within $(h, h + dw) \times (w, w + dw)$. Let $\psi_j(h, w; I) = v_j(f(hw, I), h)$, $\psi_0(0, 0; I) = v(f(0, 1), 0)$, and let $\varphi_j(h, w)dh dw$ denote the probability of choosing a job in sector j with hours of work and wage rate within $(h, h + dw) \times (w, w + dw)$. Similarly to (6) it follows that

$$\varphi_j(h, w; I) = \frac{\psi_j(h, w; I)\theta_j g_j(h, w)}{\psi_0(0, 0; I) + \sum_{k=1}^n \theta_k \int \int_D \psi_k(x, y; I) g_k(x, y) dx dy} \quad (14)$$

for $h > 0, w > 0$, and a similar formula for $h = 0, w = 0$, obtained by replacing the numerator in (14) by $\psi_0(0, 0; I)$.

If $v_j(C, h) = v(C, h)\mu_j$ where $\{\mu_j\}$ are constants, $\mu_0 = 1$, then we realize that θ_j and μ_j enter the model as $\theta_j\mu_j$, and hence θ_j and μ_j cannot be separately identified without further information or restrictions.

3. AN EMPIRICAL APPLICATION

3.1. Empirical Specification

In this application we only consider the case with two sectors, private and public. For women with a higher level of education there are more job opportunities in the public than in the private sector. Moreover, in the Norwegian public sector more emphasis has been put on facilitating combinations of work and childcare, and thus one is more likely to find a job with a subsidized day-care centre in the public than in the private sector. The public sector is more unionized than the private sector. Wages are more compressed and hours are more constrained. However, the job security tends to be higher in the public sector than in the private sector. Finally, some job

types are almost only found in the public sector (such as hospitals, colleges and universities), while others are found solely in the private sector (a large number of manufacturing firms). Thus, there are important differences between the private and the public sector that could influence the labour supply decisions of married women. Some of these differences are observed (like wages and offered hours), while others are not. The modelling framework appropriate for this application is the one outlined in Section 2.5, where sector one is the public sector and sector two is the private sector and $n = 2$.

As outlined above, we will assume that offered hours and offered wages are independent, i.e.

$$g_j(h, w) = g_{j1}(w)g_{j2}(h) \quad (15)$$

for $j = 1, 2$, where now $g_{j1}(w)$ is assumed to have all mass in one point. This means that the wage rate is only individual-specific and does not vary across jobs for a given individual and for a given sector. The density of offered hours, denoted by $g_{j2}(h)$, is assumed uniform except for peaks at full-time and part-time hours. Recall that uniformly distributed offered hours corresponds to the notion of a perfectly competitive economy. Thus, the part-time and full-time peaks in the distribution of hours capture institutional restrictions and technological constraints. We allow the sizes of the full-time and part-time peaks in the distribution of offered hours to vary across sectors. The rationale is that the public sector is more regulated than the private sector, and also that the private sector is more heterogeneous and less unionized. Thus we expect the full-time peak associated with the public sector to be higher than the full-time peak associated with the private sector. Note also that normal working hours may vary across jobs according to how strenuous the jobs are considered to be. For example, nurses, fire-workers and police officers have typically lower normal working hours than the average worker.

To facilitate estimation, we have assumed that the choice set of hours is discrete. For each sector we have specified 7 hours of work intervals. The medians of the intervals are 315, 780, 1040, 1560, 1976, 2340, and 2600 annual hours. For each sector the full-time peak occurs in the fifth interval, where the median is 1976 annual hours. The part-time peak is related to the third interval, with a median equal to 1040 annual hours. These intervals correspond to the most common agreements of what constitutes full-time and half-time annual hours of work.

In Section 2.4 we postulated particular invariance properties that allowed us to characterize the functional form of the structural part of the utility function. Unfortunately, we have not been able to provide a similar principle to characterize the functional form of $g_{j1}(w)$. Recall that $g_{j1}(w)$ is the subjective density of offered wage rates, as perceived by the agent. We shall therefore, in the present application, abandon the specification and estimation of $g_{j1}(w)$, which implies that we can only estimate the density of chosen hours of work and sector.

Assume next that

$$v_j(C, h) = v(C, h) \quad (16)$$

Hence, we can write $\psi_j(h, w; I) = \psi(h, w; I)$. Let $\bar{w}_j$ denote the mean in the offered wage rate distribution in sector j . Thus

$$\bar{w}_j = w_j^* \eta_j \quad (17)$$

for $j = 1, 2$, where $\{\eta_j\}$ are *random effects*. We assume that $\log \eta_j$, $j = 1, 2$, are independent and normally distributed, $N(0, \sigma_j)$. The systematic term, w_j^* , is assumed to vary across sectors and $\log w_j^*$ is assumed to be a linear function of length of schooling, experience and work experiences

squared. We have also experimented with random effects associated with the terms θ_j , and with the coefficient associated with leisure in the systematic part of the utility function. Thus, when accounting for the unobserved heterogeneity, it follows from (14) to (17) that the resulting choice probabilities that correspond to our observations are

$$\bar{\varphi}_j(h; w_1^*, w_2^*, I) = E \left[\frac{\psi(h, w_j^* \eta_j; I) g_{j2}(h) \theta_j}{\psi(0, 0; I) + \theta_1 \sum_{x>0} \psi(x, w_1^* \eta_1; I) g_{12}(x) + \theta_2 \sum_{x>0} \psi(x, w_2^* \eta_2; I) g_{22}(x)} \right] \quad (18)$$

for $h > 0$, $j = 1, 2$, and a similar formula for $h = 0$, obtained by replacing the numerator in (18) by $\psi(0, 0; I)$. Here the expectation is taken with respect to the random effects. In practice we compute the expectation by simulation when estimating the model.

For many reasons, most women are working in the service branch of the economy and thus for women there are more feasible jobs available in firms that provide services than elsewhere. In Norway, most of the services are provided by the public sector (health services, education, etc.) and many of the jobs here require higher education, while the services provided in the private sector, say in retail sales, are typically based on low-skill labour. Thus it is reasonable to assume that θ_j may depend on education. It is expected that the higher the education is, the higher the number of feasible jobs in the public sector. To this end we have assumed that

$$\log \theta_j = f_{j1} + f_{j2} S \quad (19)$$

where S is the length of education. We have chosen $\log v(\cdot)$ to be of the form given in (10), that is

$$\begin{aligned} \log v(C, h) = & \alpha_2 \left(\frac{[10^{-4}(C - C_0)]^{\alpha_1} - 1}{\alpha_1} \right) \\ & + \left(\frac{(L - L_0)^{\alpha_3} - 1}{\alpha_3} \right) (\alpha_4 + \alpha_5 \log A + \alpha_6 (\log A)^2 + \alpha_7 CU6 + \alpha_8 CO6) \\ & + \alpha_9 \left(\frac{[10^{-4}(C - C_0)]^{\alpha_1} - 1}{\alpha_1} \right) \left(\frac{(L - L_0)^{\alpha_3} - 1}{\alpha_3} \right) \end{aligned} \quad (20)$$

where A is the age of the married woman, $CU6$ and $CO6$ are the number of children less than 6 and above 6 years, where $L - L_0 = 1 - h/3650$, and α_j , $j = 1, 2, \dots, 8$, are unknown parameters. Observe that we have subtracted from total annual hours a 'subsistence' level, L_0 , amounting to 5110 hours, which allows for sleep and rest. This corresponds to about 14 hours per day reserved for sleep and rest.

Consistent with psychophysical evidence (Falmagne, 1985), we have also introduced a subsistence threshold level, C_0 , for consumption in the $v(\cdot)$ function. We have chosen C_0 to be close to the official estimate of a subsistence level in Norway (NOK 60 000, 1USD $\cong$ 7NOK). C is measured as the sum of the annual wage income of the woman and her husband after tax, household capital income after tax and child allowances. The tax function and the child allowances are given in appendix F of Dagsvik and Strøm (2004). If $\alpha_1 < 1$, $\alpha_3 < 1$, $\alpha_2 > 0$, and the term in front of

leisure is positive, α_9 is positive, or if negative, sufficiently small numerically, then $\log v(C, h)$ is increasing in C , decreasing in (h) for fixed C and strictly concave in (C, h) .

To facilitate the estimation procedure we have estimated the wage equation (regressed $\log w_j^*$ against the observed covariates mentioned above) in a first step by applying a version of the two-stage Heckman approach to control for selectivity, see appendix D of Dagsvik and Strøm (2004). Conditional on these estimates, the remaining parameters of the model are estimated by the maximum likelihood procedure. In the second stage only the imputed wage rates w_j^* , $j = 1, 2$, are used and the random components η_j , $j = 1, 2$, are integrated out. It would have been desirable also to allow η_1 and η_2 to be correlated. Since this will complicate the computations considerably, we have chosen to leave this challenge for another occasion. This two-stage procedure has also the advantage that it reduces the problem of negative correlation between hours of work and wage rates due to measurement error.

As regards the random effects associated with θ_1 and θ_2 , we have tried several types of specification. First we tried to specify the pair (f_{11}, f_{21}) in (19) as distributed according to a discrete binomial distribution, which can take two values with probabilities q and $1 - q$ (say). The second type of specification is based on the assumption that (f_{11}, f_{21}) is normally distributed. In addition, we have also estimated a specification where α_4 in (20) is assumed to be a normally distributed random effect.

3.2. Data

Data on the labour supply of married women in Norway used in this study consists of a merged sample from 'Survey of Income and Wealth, 1994' (Statistics Norway, 1994) and 'Level of Living Conditions, 1995' (Statistics Norway, 1995). Data covers married couples as well as cohabiting couples with common children. The age of the spouses ranges from 25 to 64. None of the spouses are self-employed and none of them are on disability or other types of benefit. All taxes paid are observed and in the assessment of disposable income, at hours not observed, all details of the tax system are accounted for. Observed hours of work are related to main job as well as possible side jobs. In 1994 the unemployment rate was rather low by international standards. Thus, for that reason it may be of little importance to employ desired hours (which we don't have) instead of actual hours. The size of the sample used in estimating the labour supply model is 824. Wage rates above NOK 350 or below NOK 40 are not utilized when estimating the wage equations. The wage rates are computed as the ratio of annual wage income to hours worked. When computing annual wage income we take into account that some women have multiple jobs. A person is working if average weekly hours of work is at least one hour. A person is a wage worker if income from self-employment is less than income from wage work.

In Table I we report the summary statistics for the sample used in estimating the labour supply model. We observe that the participation rate is rather high, at 92%. In the first place the overall participation rate among married women is very high in Norway, actually in 1994 it was the highest in the world. Second, as mentioned above, we have made a selection with respect to age, health status, etc., which contributes to increasing the participation rate further. We note that the mean wage is slightly higher in the public sector than in the private sector, whereas the wage dispersion tends to be higher in the private sector than in the public sector.

Table I. Summary statistics for married women in the sample, Norway 1994

	Not working		Public sector		Private sector	
	Mean	Std.	Mean	Std.	Mean	Std.
Age in years	40.44	9.92	41.24	8.68	38.68	9.08
Education in years	11.02	2.01	12.36	2.34	10.88	1.62
No. of children, 0–6	0.73	0.83	0.44	0.74	0.63	0.81
No. of children, 7–17	1.00	1.01	0.75	0.86	0.53	0.80
Annual hours of work	0	0	1641	489	1570	571
Disposable household income, NOK per year	322 131	200 684	329 064	122 616	331 354	125 754
Wage rate, NOK per hours			104.30	28.52	100.56	30.46
Number of observations	66		405		353	
Fractions	0.080		0.492		0.428	

Table II. Estimates of log wage equations. Single and married women, Norway 1994

Variables	Public sector		Private sector	
	Estimates	<i>t</i> -values	Estimates	<i>t</i> -values
Constant	3.37	13.5	3.70	25.2
Experience in years/100	3.21	6.0	2.55	5.1
(Experience in years/100) ²	−4.75	−5.3	−3.80	−4.2
Education in years/100	5.57	4.9	5.26	4.2
Log(probability of working in the chosen sector)	−0.12	−2.0	0.06	0.9
Variance	0.059	18.6	0.075	17.0
No. of observations	691		580	
<i>R</i> ²	0.14		0.08	

3.3. Estimation Results for the Wage Equations

In this subsection we report estimates of the wage equation. The wage equation is specified in a conventional way, i.e., the logarithm of observed wage rates, $\log w_j$, $j = 1, 2$, is assumed to depend linearly on experience, experience squared and education level. Experience is defined as age minus years of schooling minus six. To control for the possibility of selectivity we have used a simpler model than the structural model developed above. The main reason for this is that our sample used to estimate the wage equations consists of married and unmarried women, while the model above is only postulated for married women. From the simplified modelling approach adopted in this subsection it can be demonstrated that one can control for selectivity bias by applying $\log P_j$ as an additional independent variable, where P_j is the probability of being in sector j , $j = 0, 1, 2$ (where $j = 0$ means not working). The details of this approach are given in appendix D of Dagsvik and Strøm (2004). Note that if the coefficient of $\log P_j$ is negative, the conditional mean wage rate is higher than the corresponding unconditional mean. Estimates of the wage equations are given in Table II, and we observe that on the margin female workers get slightly better paid for experience and education in the public sector than in the private sector. However, the differences in returns across sectors are not significant. On a much larger sample, Barth and Røed (2001) report similar results for 1995.

Judged by R^2 , the explanatory power of the wage equations is low. Thus, it seems important to account for unobservables in the wage equation when estimating the structural model. In Section 3.1 we explained how we account for the unobservables in the wage equations. We note that the

estimated variance related to the random effects in wages tends to be higher in the private than in the public sector.

3.4. Estimates of Labour Supply Probabilities

Estimates of the parameters in the structural choice model are given in the first two number columns of Table III. Both exponents (α_1 and α_3) are significantly below 1, and the estimates imply that the deterministic part of the utility function is quasi-concave. We note that the parameter associated with the interaction term between consumption and leisure is significantly different from zero and negative. Hence, we can reject the hypothesis that the deterministic part of the utility function is additively separable in consumption and leisure.

Marginal utilities of consumption and leisure (for all relevant ages of women) are both positive. The latter depends significantly on age and number of children. Marginal utility of leisure is a convex function of age, with marginal utility of leisure increasing with age after 31–32 years of age. Marginal utility of leisure is positively affected by the number of children. We observe that the number of young and ‘old’ children has the same impact on the marginal utility of leisure. It is interesting to note that when the women are young and have children this reduces their incentive

Table III. Estimation results for the parameters of the labour supply probabilities

Variables	Uniformly distributed offered hours with part-time and full-time peaks			Uniformly distributed offered hours	
	Parameters	Estimates	<i>t</i> -values	Estimates	<i>t</i> -values
Preferences					
<i>Consumption</i>					
Exponent	α_1	0.64	7.6	0.54	6.0
Scale 10^{-4}	α_2	1.77	4.2	3.16	4.0
Subsistence level C_0 in NOK per year		60 000		60 000	
<i>Leisure</i>					
Exponent	α_3	−0.53	−2.1	−1.88	−5.1
Constant	α_4	111.66	3.2	40.92	2.5
log age	α_5	−63.61	−3.2	−22.50	−2.5
(log age) ²	α_6	9.20	3.3	3.23	2.5
# children 0–6	α_7	1.27	4.0	0.43	2.9
# children 7–17	α_8	0.97	4.1	0.30	2.7
<i>Consumption and leisure, interaction</i>					
Subsistence level of leisure in hours per year	α_9	−0.12	−2.7	−1.52	−1.9
		5120		5120	
The parameters θ_1 and θ_2; $\log \theta_j = f_{j1} + f_{j2}S$					
Constant, public sector (sector 1)	f_{11}	−4.20	−4.7	−4.87	−5.4
Constant, private sector (sector 2)	f_{21}	1.14	1.0	−0.02	−0.1
Education, public sector (sector 1)	f_{12}	0.22	2.9	0.24	3.1
Education, private sector (sector 2)	f_{22}	−0.34	−3.3	−0.26	−2.8
Opportunity density of offered hours; $g_{k2}(h)$, $k = 1, 2$					
Full-time peak, public sector (sector 1)*	$\log(g_{12}(h_{\text{Full}})/g_{12}(h_0))$	1.58	11.8		
Full-time peak, private sector (sector 2)	$\log(g_{22}(h_{\text{Full}})/g_{22}(h_0))$	1.06	7.4		
Part-time peak, public sector	$\log(g_{12}(h_{\text{Part}})/g_{12}(h_0))$	0.68	4.4		
Part-time peak, private sector	$\log(g_{22}(h_{\text{Part}})/g_{22}(h_0))$	0.80	5.2		
# observations		824		824	
Log-likelihood		−1760.9		−1862.0	

* The notation h_0 refers to an arbitrary level of hours of work different from full-time and part-time hours.

to participate in labour market activities, and when they are older and without children the age effects gradually also reduce their incentive to participate in labour market activities.

The exponent α_1 is significantly different from zero, which means that agents do not only care about relative consumption levels (beyond subsistence). Absolute levels also matter. The exponent α_3 is almost not significantly different from zero (at a 5% level or less), which means that the deterministic part of the utility function is close to being log-linear in leisure.

The estimates of the opportunity density confirm the conjecture that there are more jobs available in the public sector for higher educated women than for women with lower education. This means that if length of schooling is increased while wage rates are kept fixed, participation in the public sector will increase. In the private sector education does seem to have the opposite effect. At first glance this seems to indicate that highly educated women tend to be ‘overqualified’ in the private sector. However, this may also be due to a preference effect: if preferences depend on schooling, such that highly educated women tend to prefer jobs in the public sector (university, hospitals), this would imply that a corresponding preference term is decreasing in S , cf. section 2.5.

Moreover, the full-time peak is more distinct in the public sector than in the private sector. As mentioned above, this may be due to the fact that the public sector is more unionized than the private one.

It turned out to be very difficult to estimate the random effects associated with θ_1 and θ_2 . We have tried a discrete binomial and the normally distributed random effects specifications, and both yielded degenerate results. Also, we were not able to estimate specifications based on a discrete distribution with more than two support points. Although we cannot rule out the possibility that other types of random effect may exist, we nevertheless think our estimation result indicates that it is likely that any random effect is negligible. Analogous to van Soest *et al.* (2002), we have also tried to include a random effect in the leisure term of the utility function. Similarly to van Soest *et al.*, we find that the t -value of this random effect is practically equal to zero.

Table IV compares observed and predicted aggregates, and in Figure 1 we show the observed fractions and predicted probabilities across the different hours of work categories (category 1 is not working, categories 2–8 are public sector and categories 9–15 are private sector). We note that our model predicts the observed frequencies quite well.

As noted above, we can in general only identify the product $v(C, h)g_{j2}(h)$ non-parametrically. To disentangle $v_2(h)$ from $g_2(h)$ we have assumed that the clustering of hours of work at part-time and full-time work is due to technological organizational constraints and/or regulation of hours introduced by unions and/or the government. The term $g_{j2}(h)$ is meant to capture this aspect of the labour market in the highly unionized Norwegian economy. Thus, through parametric identification our model implies that observed concentration of hours of work around part-time and full-time work arises because there are institutional constraints in the labour market rather than because

Table IV. Observed and predicted aggregates, married women, Norway 1994

Variables	Not working		Public sector		Private sector	
	Observed	Predicted	Observed	Predicted	Observed	Predicted
Choice probabilities	0.080	0.079	0.492	0.483	0.428	0.438
Annual hours	0	0	1641	1585	1570	1632
McFadden's ρ^2				0.211		
McFadden's adjusted ρ^2				0.195		

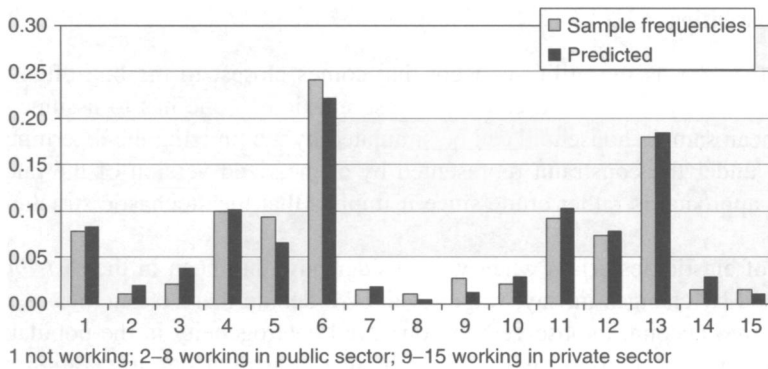


Figure 1. Probabilities and frequencies

individuals have strong preferences for full-time and part-time hours of work. If the parameters of the utility function are robust with respect to our assumption, then our empirical model may also be applied to simulate the impact of a change in the institutional constraints on available working hours in the market.

To contrast our approach with the more familiar one with uniformly distributed offered hours, we have re-estimated the model under the assumption that *offered* hours is uniformly distributed. It goes without saying that the parameters attached to the leisure term will be affected when we force the clustering of hours to be explained solely by preferences. Of greater interest is how the estimates of the other parameters of the utility function are affected when offered hours are assumed to be uniformly distributed. In the third and fourth columns of Table III the estimates for this case are reported. We note that the parameters in the sector-specific opportunity densities remain unchanged. The exponent of the leisure term is significantly different from zero and negative when offered hours are uniformly distributed. The numerical value is higher when offered hours are uniformly distributed compared to the case with full-time and part-time peaks in offered hours. When the clustering of hours is due to spikes in the distribution of offered hours, labour supply responses to marginal wage changes become weak, given a full-time or part-time job. But in this case the Frisch elasticity, $(1/(1-\alpha_3))$, that can be derived from the estimates of preferences, is rather large. This means that if the constraints on offered hours are removed, labour supply responses to wage changes will increase. If it is assumed that offered hours are uniformly distributed and the model is estimated on data in which offered hours are constrained, then the resulting weak labour supply responses will be interpreted as due to preferences. Moreover, if it is wrongly assumed that offered hours are uniformly distributed, and a labour supply model is estimated on two different data sets, one with and one without constraints on offered hours, the estimate of α_3 will differ and the difference will wrongly be interpreted as a shift in preferences.

Second, we have, for the sake of goodness of fit comparisons, also estimated conventional reduced form models for supplied hours of work, conditional on sector. The reduced form hours of work equations are specified as linear functions of length of education, age, age squared, number of children below six years, number of children above six years and log of disposable non-labour income and normally distributed error terms. To be able to compare the fit with the structural model above we have only used the discretized observations (seven intervals with the same partition as above). Observationally, this yields sector-specific empirical models that have the ordered probit structure. Measured by McFadden's ρ^2 the fit is bad, less than 0.02 for both models.

3.5. Wage Elasticities

The mean utility, $\psi(\cdot)$, is the utility concept that comes closest to the one often used by others in the calculation of elasticities. To calculate these elasticities one has to assume that the labour supply of the mean sample household can be simulated by maximizing the deterministic part of the utility function under the constraint represented by a linearized version of the budget constraint. Of course, this approach is rather crude since it implies that the stochastic structure of the model is ignored.

Another set of elasticities arises when we consider how the mean in the *distribution* of labour supply is affected by changes in, say, wage levels. We denote these elasticities as *aggregate* ones since they take into account unobserved and observed heterogeneity in the population. Moreover, they also account for the non-convexity of the budget constraint due to taxation and restrictions on hours, and are thus consistent with the structure of the model.

In Table V we report what we call aggregate uncompensated elasticities. They are calculated as follows: the model is used to simulate the labour supply for each female under the current regime and when the wage rates in each sector, and in both sectors, respectively, are increased by 1%. The aggregate elasticity of female labour supply is obtained by calculating the relative change in the mean female labour supply (over all females in the sample) that results from a 1% wage increase for the females, *ceteris paribus*. Elasticities are numerically small, apart from the case when sector-specific wage levels are changed. The reasons why overall responses are small are due to high overall labour market participation among married Norwegian women and a regulated and rigid market of offered hours of work. However, we note that when the wage level in the public sector is increased by 1%, the participation probability in this sector increases by as much as 1.52%. Most of this increase comes at the expense of a decrease in the participation probability in the private sector of -1.29% . Overall participation increases by a minor 0.15%. A similar pattern emerges when the wage level in the private sector is raised by 1%. The probability of working in the private sector increases by 1.80% and as in the preceding case, most of the increase comes from a reduction in the probability of working in the other sector, the public sector (-1.40%). A similar pattern emerges for annual hours of work, conditional on sector, but the impact is smaller. From the last row in Table V we notice that an overall increase of 1% is estimated to increase supplied hours in our sample by 0.65%, with an

Table V. Aggregate uncompensated wage elasticities, married women, Norway 1994

Variables	Choice probabilities and mean hours before wage changes	Elasticities with respect to changes in wage rates		
		Public sector	Private sector	Both sectors
Participation probabilities				
Working	0.921	0.15	0.16	0.28
Working in public sector	0.483	1.52	−1.40	0.09
Working in private sector	0.438	−1.29	1.80	0.48
Mean annual hours of work, conditional on sector				
Public sector	1585	0.36	0.03	0.37
Private sector	1632	−0.09	0.17	0.34
Mean total annual hours of work				
Unconditional	1480	0.34	0.37	0.65

almost equal split between increased participation and increased supply of hours, conditional on working.

3.6. A Flat Tax Reform

In Table VI we show the result of replacing the tax system of 1994 by a flat tax rate, which is determined so as to keep the tax revenue (for married couples) unchanged. In doing so, we account for the fact that the change in the tax system will also change after-tax wage income of the husbands. The tax of 28% on capital income is kept constant. We find that a tax rate of approximately 29% charged from the very first Krone earned brings in the same tax revenue as the 1994 tax rules do. It is interesting to note that this tax rate is close to the tax rate charged on capital income. From Table VI we note that this potential tax reform increases overall labour supply, participation as well as hours worked, but not by much. The overall participation probability increases by merely 0.07 percentage points, that is, from 92.1% to 92.8%. Expected hours of work in the total population is increased by 108 hours, altering annual hours of work from 1480 to 1588. These changes shadow for stronger responses across sectors and deciles in the wage distribution.

The flat tax reform implies a flow of workers, away from the public sector and towards the private sector. The overall probability of working in the public sector is reduced from 48.3% to 45.6%, whereas the private sector participation probability increases from 43.8% to 46.5%. These flows increase with the wage rate. Women who belong to the top decile are predicted to increase their participation probability in the private sector by as much as 7.56 percentage points, that is, from 24.7% to 32.3%. Most of this increase is due to a reduction among 'well-paid' women working in the public sector. The reasons for these responses are first, that this tax reform forces women in the lowest deciles to work harder (they have to pay higher taxes on the margin as well as a percentage of total income). Second, the women at the top of the wage rate distribution get a substantial reduction in the marginal tax rates. This incentive to work more dominates over the fact that the average tax, taxes paid relative to household income, is also reduced. Our calculations of income elasticities (not shown here) indicate that the income effects are not strong.

The women in the middle of the wage rate distribution get almost no change in their marginal tax rate, but they enjoy a reduction in taxes paid relative to household income (remember that the tax rate on the labour income of their spouse is also changed). As shown in Table VI, there is a

Table VI. A flat tax reform with the same tax revenue as with the 1994 tax rules. Percentage points changes in probabilities and changes in annual hours. Flat tax relative to 1994 rules. The flat tax rate is 29%

Variables	Women's wage, first deciles	Women's wage, middle deciles	Women's wage, top deciles	All
Percentage points				
<i>Probability of working</i>	0.57	-0.14	1.13	0.07
Probability of working in public sector	-0.87	-2.44	-6.43	-2.65
Probability of working in private sector	1.44	2.30	7.56	2.72
Annual hours (% changes)				
Mean annual hours of work conditional on sector				
<i>Public sector</i>	46 (3.1)	80 (5.1)	118 (7.2)	80 (5.1)
Private sector	99 (6.5)	148 (9.2)	216 (12.2)	126 (7.8)
Mean total annual hours of work, unconditional	76 (5.9)	105 (7.2)	169 (10.4)	108 (7.4)

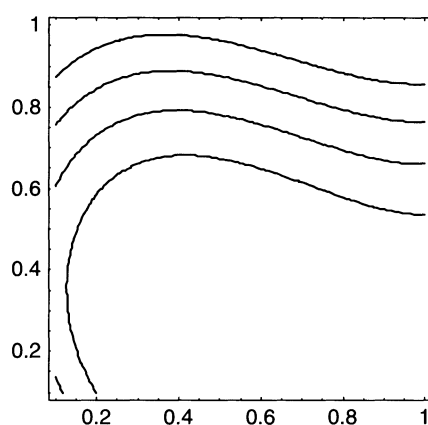


Figure 2. Third-order polynomial utility function. Consumption along the vertical axis and leisure along the horizontal axis

minor reduction in the probability of working among women in the middle deciles. Finally, a flat tax of 29% makes it more attractive to increase hours of work, in particular among the women with the highest earning potential and also, especially in the private sector, where the regulation of hours is less than in the public sector.

3.7. Comparison Between our Model and a van Soest-Type Model Specification

In this subsection we discuss the results of estimating the flexible preference model (FPM) of van Soest *et al.* (2002). The estimates are reported in appendix E of Dagsvik and Strøm (2004). First we have estimated a specification where the utility function is assumed to be a polynomial of degree 5. A representation of fixed costs of working (which is unobserved) is included as well as a random effect associated with leisure in a completely similar way as in van Soest *et al.* (2002). The random effect in the wage equation is, however, included in the same way as in our model described above. From Table E.1 in Dagsvik and Strøm (2004) it is found that most of the coefficients are not sharply determined, with the exception of the coefficients attached to the variables that are assumed to affect the fixed costs of working. Similarly to van Soest *et al.* (2002) we obtain the unreasonable result that fixed cost of working decreases with the number of children. By comparing the likelihood values of Table E.1 and Table III it is clear that our model explains data much better than the FPM. Recall that our model differs from the FPM not only in the specification of the functional form of the utility function, but also in the specification of the opportunity sets. A very disturbing fact with the estimates of the FPM reported in Table E.1 of Dagsvik and Strøm (2004) is that for most households fixed costs of working exceed after-tax wage income, given positive hours, which yields negative disposable household incomes among married women.

Second, we have estimated the FPM without a representation of fixed costs of working. The empirical estimates for third- and fifth-order polynomial utility specifications are given in Table E.2 of Dagsvik and Strøm (2004). Again, our model explains data much better than the van Soest *et al.* (2002) model. In Figures 2 and 3 we give the estimated indifference curves for an average woman and observe that the estimated utility function is not quasi-concave.

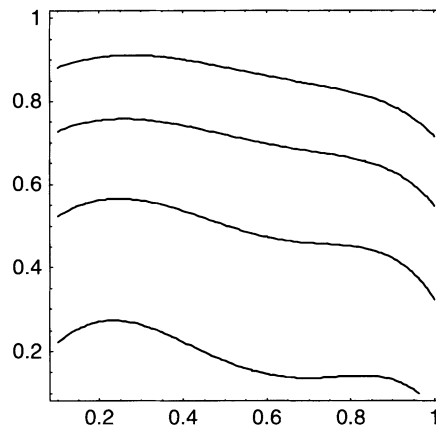


Figure 3. Fifth-order polynomial utility function. Consumption along the vertical axis and leisure along the horizontal axis

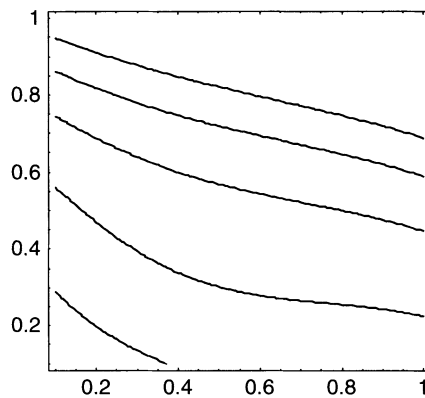


Figure 4. Third-order polynomial utility function, combined with our specification of opportunities. Consumption along the vertical axis and leisure along the horizontal axis

Finally, in Table E.3 of Dagsvik and Strøm (2004), we report estimates based on combining a third-order polynomial utility specification with our specification of opportunity sets. Measured by McFadden's adjusted ρ^2 (see Ben-Akiva and Lerman, 1985), the goodness of fit of this combined model is the same as for our model, and Figure 4 also indicates that the utility function is quasi-concave in the central area of consumption and leisure combinations. However, from Figure 4 we note that the indifference curves for the polynomial utility specification imply the unreasonable property that at high consumption levels the needed compensation (in terms of consumption) for working one more hour is almost the same irrespective of hours worked. In contrast, the corresponding indifference curves for the Box–Cox utility specification displayed in Figure 5 show a more reasonable pattern, namely that the needed compensation for working one more hour is sharply increasing with hours of work for high levels of hours of work.

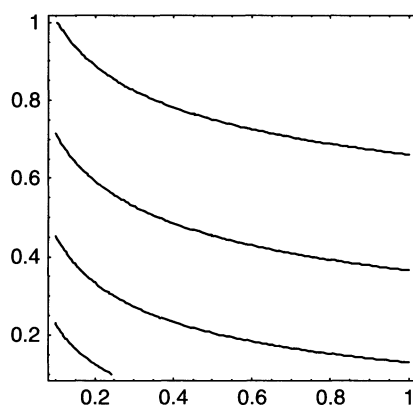


Figure 5. Box–Cox utility function, with our specification of opportunities. Consumption along the vertical axis and leisure along the horizontal axis

5. CONCLUSION

In this paper we have discussed a particular approach for labour supply modelling, with special reference to the following issues: (i) households may have preferences over jobs, characterized by job- and sector-specific non-pecuniary attributes, hours of work and wage rates; (ii) exact representation of complicated and non-convex budget constraints, and convenient representations of the set of feasible job attributes; (iii) justification of the functional form of the utility function and the distribution of unobserved taste-shifters. We have demonstrated that the framework presented proves to be practical for dealing with (i) and (ii). A more fundamental theoretical issue is the problem of characterizing the functional form of the empirical model on the basis of theoretical principles. By drawing on the recent literature in measurement theory and theoretical psychophysics, we have shown that it is possible to apply invariance principles to constrain and justify the class of admissible functional forms for the utility function.

An empirical version of the model has been estimated on a sample of Norwegian married women. The estimated model turns out to reproduce the data quite well. It is demonstrated that it is of empirical importance to distinguish between job opportunities across sectors of the economy. We also report the impact on labour supply of replacing the 1994 tax rules by a revenue-neutral flat tax reform.

We have also investigated the modelling framework with flexible preferences (FPM), proposed by van Soest *et al.* (2002). Although more experience with that framework is needed before a definite conclusion can be reached, the empirical results we have obtained indicate that the FPM framework does not seem to be promising as a general research strategy for labour supply analysis.

Although we have argued for an alternative approach to the modelling of structural labour supply in this paper, we must admit that it is currently too demanding to carry out a fully satisfactory analysis. Some of the difficulties we face are due to lack of data, like more job-specific observable quantitative attributes or indicators. Other limitations are due to the modelling framework. For example, it is so far not clear how one should make a structural specification of the opportunity density. More generally, it is not clear what kind of information it is reasonable to assume that the agents have about their choice sets, and to what extent they make decisions in an intertemporal setting. Surely, these issues are also important concerns for all labour supply analyses. They

may, however, be even more crucial for models such as the one proposed in this paper, where representations of latent choice constraints and preferences over latent qualitative opportunities play an important role.

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REFERENCES

- Aaberge R, Dagsvik JK, Strøm S. 1990. Labor supply, income distribution and excess burden of personal income taxation in Sweden. Discussion Paper No. 53, Statistics Norway.
- Aaberge R, Dagsvik JK, Strøm S. 1995. Labor supply responses and welfare effects of tax reforms. *Scandinavian Journal of Economics* **97**: 635–659.
- Aaberge R, Colombino U, Strøm S. 1999. Labor supply in Italy: an empirical analysis of joint household decisions, with taxes and quantity constraints. *Journal of Applied Econometrics* **14**: 403–422.
- Altonji JG, Paxson CH. 1988. Labor supply preferences, hours constraints and hours–wage trade-offs. *Journal of Labor Economics* **6**: 254–276.
- Anderson I, Dagsvik JK, Strøm S, Wennemo T. 1988. Non-convex budget sets, hours restrictions and labor supply in Sweden. Discussion Papers, Statistics Norway.
- Arrufat JL, Zabalza A. 1986. Female labor supply with taxation, random preferences and optimization errors. *Econometrica* **54**: 47–63.
- Barth E, Røed M. 2001. Norway. In *Education and Earnings in Europe*, Harmon C, Walker I, Westergaard-Nielsen N (eds). Edward Elgar: Cheltenham, UK; pp. 198–212.
- Ben-Akiva M, Lerman SR. 1985. *Discrete Choice Analysis*. MIT Press: Cambridge, MA.
- Blomquist S. 1983. The effect of income taxation on the labor supply of married men in Sweden. *Journal of Public Economics* **17**: 169–197.
- Blomquist S. 1992. Estimation methods for male labor supply functions. How to take account of nonlinear taxes. *Journal of Econometrics* **70**: 383–405.
- Blundell R, MaCurdy T. 1999. Labor supply: a review of alternative approaches. In *Handbook of Labor Economics*, Ashenfelter O, Card D (eds). North Holland: New York.
- Burtless G, Hausman JA. 1978. The effects of taxation on labor supply. *Journal of Political Economy* **86**: 1103–1130.
- Copeland A, Monnet C. 2004. *The welfare effects of incentive schemes*. <http://www.marginalq.com/Copeland>
- Dagsvik JK. 1994. Discrete and continuous choice, max-stable processes and independence from irrelevant attributes. *Econometrica* **62**: 1179–1205.
- Dagsvik JK. 2000. Aggregation in matching markets. *International Economic Review* **41**: 27–57.
- Dagsvik JK, Strøm S. 1997. A framework for labor supply analysis in the presence of complicated budget restrictions and qualitative opportunity aspects. Memorandum from the Department of Economics, University of Oslo, No. 22.
- Dagsvik JK, Strøm S. 2004. Sectoral labor supply, choice restrictions and functional form. Discussion papers, no. 388, Statistics Norway. <http://www.ssb.no/publikasjoner/Dp/pdf/dp388.pdf>
- Dickens W, Lundberg S. 1993. Hours restrictions and labor supply. *International Economic Review* **34**: 169–191.
- Edin P-A, Zetterberg J. 1992. Inter-industry wage differentials: evidence from Sweden and a comparison with the United States. *American Economic Review* **82**: 1341–1349.
- Falmagne JC. 1985. *Elements of Psychophysical Theory*. Oxford University Press: New York.
- Hausman JA. 1980. The effects of wages, taxes and fixed costs on women's labor force participation. *Journal of Public Economics* **14**: 161–192.

- Hausman JA. 1981. Labor supply. In *How Taxes Affect Behavior*, Aaron H, Pechman J (eds). Brookings Institution: Washington, DC.
- Hausman JA. 1985. The econometrics of non-linear budget sets. *Econometrica* **53**: 1255–1282.
- Hausman JA, Ruud P. 1984. Family labor supply with taxes. *American Economic Review* **74**: 242–253.
- Heim BT, Meyer BD. 2003. Structural labor supply models when budget constraints are non-linear. Working paper, The Harris School of Public Policy, University of Chicago.
- Ilmakunnas S, Pudney S. 1990. A model of female labour supply in the presence of hours restrictions. *Journal of Public Economics* **41**: 183–210.
- Kapteyn A, Kooreman P, van Soest A. 1990. Quantity rationing and concavity in a flexible household labor supply model. *The Review of Economics and Statistics* **62**: 55–62.
- Krueger AB, Summers LH. 1988. Efficiency wages and the inter-industry wage structure. *Econometrica* **56**: 259–293.
- Luce RD. 1959. *Individual Choice Behavior*. John Wiley & Sons: New York.
- MaCurdy T, Green D, Paarsch H. 1990. Assessing empirical approaches for analyzing taxes and labor supply. *Journal of Human Resources* **25**: 415–490.
- MaCurdy T. 1992. Work disincentive effects of taxes: a reexamination of some evidence. *American Economic Review, Papers and Proceedings* **82**: 243–249.
- McFadden D. 1984. Econometric analysis of qualitative response models. In *Handbook of Econometrics*, Vol. II, Griliches Z, Intriligator MD (eds). Elsevier Science: New York.
- Moffitt R. 1984. The estimation of a joint wage hours labor supply model. *Journal of Labor Economics* **2**: 550–566.
- Resnick S. 1987. *Extreme Values, Regular Variation and Point Processes*. Springer-Verlag: New York.
- Statistics Norway. 1994. Survey of Income and Wealth. www.ssb.no/emner/05/01/
- Statistics Norway. 1995. Level of Living Conditions. www.ssb.no/emner/00/02/
- Thurstone LL. 1927. A law of comparative judgment. *Psychological Review* **34**: 273–286.
- Strauss D. 1979. Some results on random utility models. *Journal of Mathematical Psychology* **17**: 32–35.
- Van Soest A. 1994. Structural models of family labor supply. A discrete choice approach. *The Journal of Human Resources* **XXX**: 63–88.
- Van Soest A, Das M, Gong X. 2002. A structural labour supply model with flexible preferences. *Journal of Econometrics* **107**: 345–374.